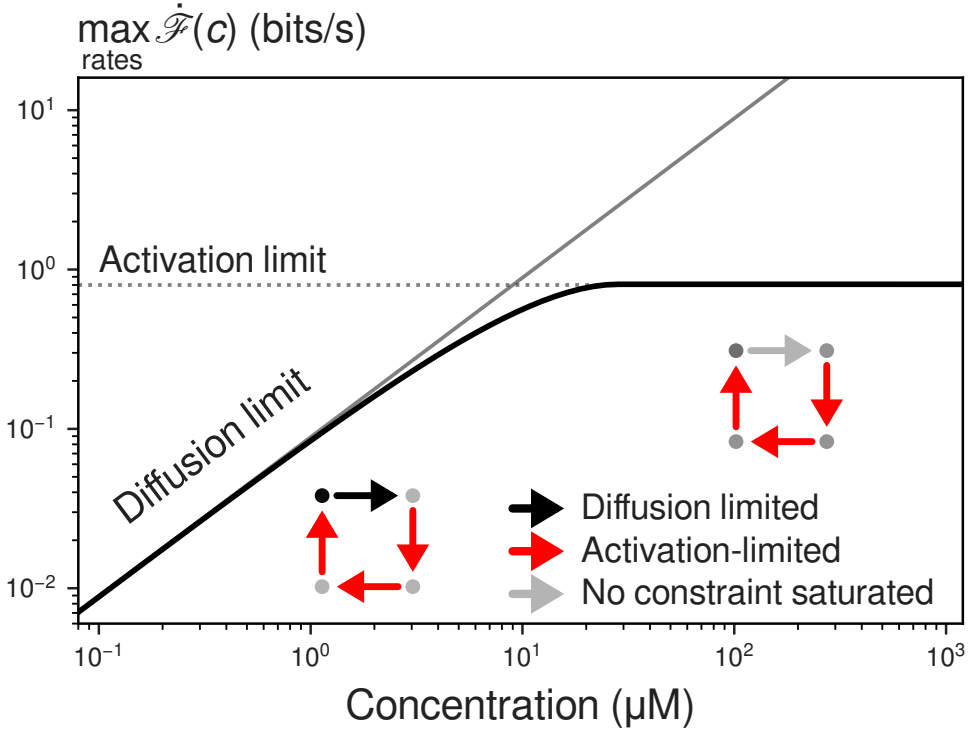
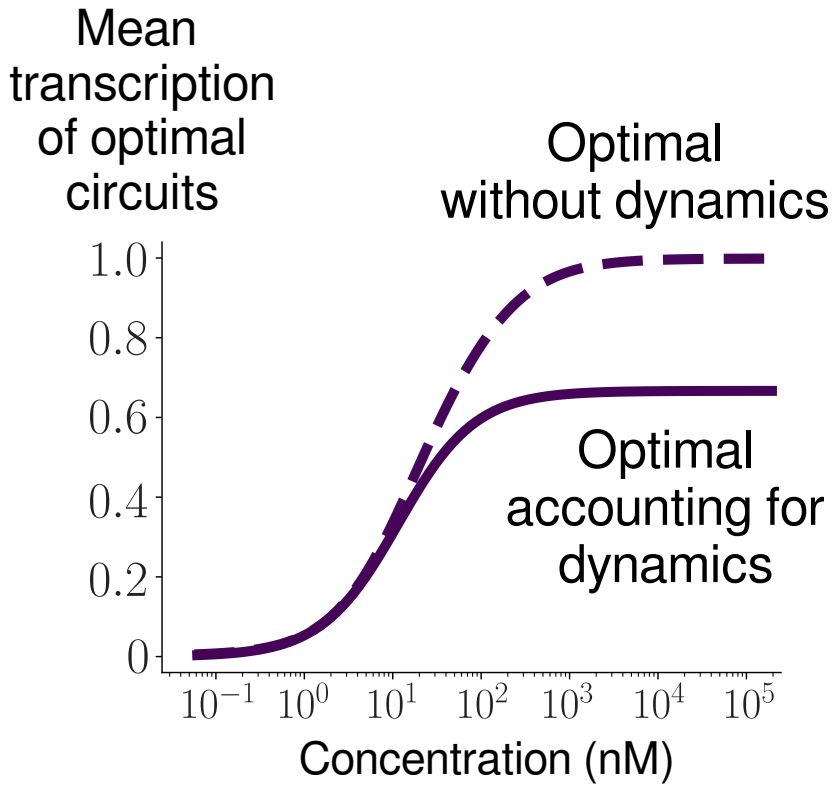

max $\dot{\mathcal{F}}(c)$ (bits/s)
rates
10^1
10^0
10^{-1}
10^{-2}
Activation limit
Diffusion limit
Diffusion limited
Activation-limited
No constraint saturated
10^{-1}
10^0
10^1
10^2
10^3
Concentration (μM)


Mean transcription of optimal circuits
Optimal without dynamics
Optimal accounting for dynamics
1.0
0.8
0.6
0.4
0.2
0
10^{-1}
10^0
10^1
10^2
10^3
10^4
10^5
Concentration (nM)